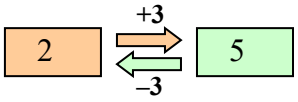
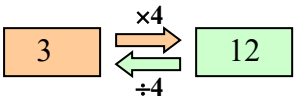


Indefinite Integration (不定積分)

There are lots of inverse operations (逆運算) in mathematics, for example





And the inverse operation of Differentiation is Integration (積分) :

Differentiation 微分

$$\frac{d}{dx}(x^2) = 2x$$

$$\frac{d}{dx}(x^2+1) = 2x$$

$$\vdots$$

$$\frac{d}{dx}(x^2+C) = 2x$$

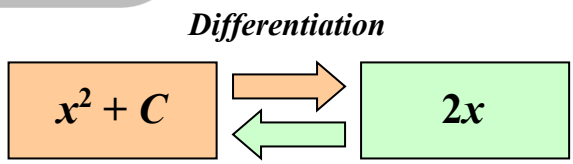


Integration 積分

$$\int 2x\,dx = x^2 + C$$

where C is constant

i.e.



Since the value of C is undetermined (不定), the above is called an indefinite integral. C is referred to as the ***integration constant*** **(積分常數)**.

The following is the definition of an indefinite integral:

Consider a function $f(x)$. If $F(x)$ is a function such that

$$\frac{d}{dx}F(x) = f(x)$$

then the indefinite integral (不定積分) of $f(x)$ is defined as

$$\int f(x)dx = F(x) + C$$

where C is constant (it can be found if more information is given)

Not for sale

Rules of Integration (Power Rule)

Rule No. 1 Power Rule 冪法則 (For $n \neq -1$)

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C \quad \text{where } \boldsymbol{C} \text{ is constant}$$

e.g. ***1.*** $\int x \, dx = \frac{x^2}{2} + C$

2. $\int x^2 \, dx = \frac{x^3}{3} + C$



3. $\int x^3 \, dx =$

4. $\int x^5 \, dx =$

5. $\int x^{10} \, dx =$

6. $\int x^{-2} \, dx =$

7. $\int x^{-4} \, dx =$

8. $\int x^{-7} \, dx =$

9. $\int x^{\frac{1}{2}} \, dx =$

10. $\int x^{\frac{3}{2}} \, dx =$

Not for sale

Rules of Integration (Constant Rule)

Rule No. 2 : **Constant Rule (*k* is a constant 常數)**

$$\int k\ dx = kx + C$$

e.g. ***1.*** $\int 3\ dx = 3x + C$



2. $\int dx =$

Rule No. 3 : **Constant Multiple Rule (*k* is a constant)**

$$\int k \cdot f(x)\ dx = k \cdot \int f(x)\ dx$$

e.g. ***1.*** $\int 2x^4\ dx = 2 \cdot \frac{x^5}{5} + C = \frac{2}{5}x^5 + C$



2. $\int 5x^7\ dx =$

3. $\int \frac{1}{2}x^{10}\ dx =$

4. $\int 3x^{-3}\ dx =$

5. $\int 3x^{-\frac{1}{3}}\ dx =$

Not for sale



Rules of Integration (Sum and Different Rule)

Rule No. 4 : **Sum and Different Rule** 加及減法則

$$\int [f(x) \pm g(x)]\,dx = \int f(x)\,dx \pm \int g(x)\,dx$$

e.g. ***1.*** $\int (x^2+1)\,dx = \frac{x^3}{3} + x + C$



2. $\int (3x^2-1)\,dx =$

3. $\int (2x^2-3x+4)\,dx =$

4. $\int (x^{-2}-3x^{-5})\,dx =$

5. $\int (x+\sqrt{x})\,dx =$

6. $\int (x+1)(x+2)\,dx =$

7. $\int \frac{x^4+1}{x^2}\,dx =$

Not for sale

Rules of Integration

Rule No. 5 : $\underline{\text{For } x \neq 0}$

$$\int \frac{1}{x}\,dx = \ln|x| + C$$

e.g. ***1.*** $\int \frac{6}{x}\,dx = 6\ln|x| + C$



2. $\int \frac{1}{2x}\,dx =$

Rule No. 6 :

$$\int e^{kx}\,dx = \frac{1}{k}e^{kx} + C$$

e.g. ***1.*** $\int e^{2x}\,dx = \frac{1}{2}e^{2x} + C$



2. $\int \frac{1}{e^{2x}}\,dx =$

3. $\int e^{\frac{x}{3}}\,dx =$

4. $\int e^{4x+2}\,dx =$

Not for sale

Rules of Integration : Integration by substitution

When calculating integrals of composite functions, using integration by substitution (代換積分法) is a powerful and effective method!

e.g. Evaluate $\int (2x+1)^{100} dx$.

Let $u = 2x + 1$

設 u 為括號裏的函數

Then $\frac{du}{dx} = 2$, i.e. $du = 2dx$

將 du 以 dx 表示

Therefore

$$\int (2x+1)^{100}\, dx = \frac{1}{2}\int (2x+1)^{100} \cdot 2dx$$

$$= \frac{1}{2}\int u^{100} du$$

將 $2x + 1$ 換成 u
及將 $2dx$ 換成 du

$$= \frac{1}{2} \cdot \frac{u^{101}}{101} + C$$

$$= \frac{1}{202}(2x+1)^{101} + C$$

計算積分，並將答案以 x 表示

Try it: Show that $\int (ax+b)^n dx = \frac{1}{a} \cdot \frac{(ax+b)^{n+1}}{(n+1)} + C$.

Not for sale

Rules of Integration : Integration by substitution



[A.Math 2005 Q.1]

Find $\int (2x-7)^7 \, dx$. ***(2 marks)***

[A.Math 2004 Q.1(b)]

Find $\int (2-x)^{2004} \, dx$. ***(2 marks)***

Not for sale

Caution

Rules of Integration : Integration by substitution

There is no product rule for integration, so to integrate composite functions involving multiplication, such as

Evaluate $\int x(2x+1)^{100}dx$.

We will then apply the substitution method for integration.

e.g. Evaluate $\int x(2x+1)^{100}dx$.

Let $u = 2x + 1$

設 u 為括號裏的函數

Then $\frac{du}{dx} = 2$, i.e. $du = 2dx$

將 du 以 dx 表示

Therefore

$$\int x(2x+1)^{100}\,dx = \frac{1}{2}\int (x)(2x+1)^{100}\cdot 2dx$$

$$= \frac{1}{2}\int \left(\frac{u-1}{2}\right)\cdot u^{100}du$$

將 x 換成 $\frac{u-1}{2}$
及將 $2x+1$ 換成 u
及將 $2dx$ 換成 du

$$= \frac{1}{4}\int (u^{101} - u^{100})du$$

將括號展開，
純以加或減來表示算式!

$$= \frac{1}{4}\cdot\frac{u^{102}}{102} - \frac{1}{4}\cdot\frac{u^{101}}{101} + C$$

$$= \frac{u^{102}}{408} - \frac{u^{101}}{404} + C$$

$$= \frac{(2x+1)^{102}}{408} - \frac{(2x+1)^{101}}{404} + C$$

計算積分，並將答案以 x 表示

Not for sale

Rules of Integration : Integration by substitution



[A.Math 1999 Paper II Q.2]

Find $\int x(x-2)^{99}\,dx$. ***(4 marks)***

[A.Math 1997 Paper II Q.2]

Find $\int x\sqrt{x-1}dx$. ***(4 marks)***

Not for sale

Caution

Rules of Integration : Integration by substitution

Similarly, integration does not have a quotient rule, so when integrating functions involving division such as

Evaluate $\int \frac{x}{x^2+1}\,dx$.

We will then apply the substitution method for integration.

e.g. Evaluate $\int \frac{x}{x^2+1}\,dx$.

Let $u = x^2 + 1$ — 設 u 為分母中的函數

Then $\frac{du}{dx} = 2x$, i.e. $du = 2x\,dx$ — 將 du 以 dx 表示

Therefore

$$\int \frac{x}{x^2+1}\,dx = \frac{1}{2}\int \frac{1}{x^2+1}\cdot 2x\,dx$$

$$= \frac{1}{2}\int \frac{1}{u}\,du$$ — 將 $x^2 + 1$ 換成 u 及將 $2xdx$ 換成 du

$$= \frac{1}{2}\ln|u| + C$$

$$= \frac{1}{2}\ln|x^2+1| + C$$ — 計算積分，並將答案以 x 表示

Note: 無時無刻也請記住，我們可以將加減數分開來積分：

$$\int [f(x) \pm g(x)]\,dx = \int f(x)\,dx \pm \int g(x)\,dx$$

但**絕不能將乘除數分開來積分**!!

$$\int [f(x) \times g(x)]\,dx \neq \int f(x)\,dx \times \int g(x)\,dx$$

Not for sale



Rules of Integration : Integration by substitution



[M2 2012 Q.4]

(a) Find $\int \frac{x+1}{x} dx$.

(b) Using the substitution $u = x^2 - 1$, find $\int \frac{x^3}{x^2 - 1} dx$. ***(5 marks)***

Not for sale



Rules of Integration : Integration by substitution

Choosing the correct substitution is the crucial first step in integration by substitution! Can you successfully take this step?

Original expression	Substitution: let u =	Relation between du and dx	New expression
1. $\int \frac{1}{2x+1}\,dx$	$u = 2x + 1$	$du = 2dx$	$\frac{1}{2}\int \frac{1}{u}\,du$
2. $\int (3x+2)^7\,dx$			
3. $\int \sqrt{2x+1}\,dx$			
4. $\int x(1-x^2)^8\,dx$			
5. $\int x^2\sqrt{x+2}\,dx$			
6. $\int \frac{1}{(2x+1)^6}\,dx$			
7. $\int \frac{4x-2}{\sqrt{x^2-x+3}}\,dx$			
8. $\int e^{4x-2}\,dx$			
9. $\int xe^{x^2}\,dx$			
10. $\int \frac{4e^{2x}}{e^{2x}+1}\,dx$			

Not for sale

Rules of Integration : Integration by substitution

[P.Math 2006 Paper II Q.4(a)]

Using the substitution $t = \sqrt{1+x^2}$, find $\int \frac{x^3}{\sqrt{1+x^2}} dx$. ***(3 marks)***

(**Remark:** After completing the question, try to substitute $u = 1+x^2$ and solve the problem again to justify which substitution is better.)

Not for sale

Application of Integration

Two major application of Indefinite Integration and their characteristics:

1. **From given slope (e.g. $\frac{dy}{dx}$=……) to find equation of a curve (*y* = ……)**

   - 這些題目會首先給出 $\frac{dy}{dx}$=…… 或 f'(*x*) = ……. ，並且大多數會註明此算式代表的是斜率 (slope)
   - 題目最後求曲線的方程 (equation of the curve)

2. **From given rate of change (e.g.: $\frac{dy}{dt}$=……) to find original function (*y* =…)**

   - 這些題目會首先給出 $\frac{dy}{dt}$=…… 或 f'(*t*) = ……. ，並且大多數會註明此算式代表的是變率 (rate of change)
   - 題目最後要以 *t* 表示 *y* (express *y* in terms of *t*) 或求 f(t)

e.g. The slope at any point (x, y) of a curve is $x^2 - 3x + 1$. If the curve passes through $(0, 2)$, find the equation of the curve.

$$\frac{dy}{dx} = x^2 - 3x + 1$$

$$y = \int (x^2 - 3x + 1)dx$$

$$= \frac{1}{3}x^3 - \frac{3}{2}x^2 + x + C$$

用積分求得曲線方程

Substitute $(0, 2)$ into $y = \frac{1}{3}x^3 - \frac{3}{2}x^2 + x + C$,

$$2 = \frac{1}{3}(0)^3 - \frac{3}{2}(0)^2 + 0 + C \qquad \therefore C = 2$$

用代入法求積分常數

$\therefore$ The equation of the curve is $y = \frac{1}{3}x^3 - \frac{3}{2}x^2 + x + 2$.

Not for sale

Application of Integration



[A.Math 2004 Q.4]

The slope at any point (x, y) of a curve C is given by $\frac{dy}{dx} = 3x^2 + 1$. If the x-intercept of C is 1, find the equation of C. ***(4 marks)***

Not for sale

Application of Integration



[A.Math 1987 Paper II Q.5]

The slope of at any point (x, y) of a curve is given by

$$\frac{dy}{dx} = (3x^2 - 2)(x^3 - 2x + 1)^{\frac{1}{3}}$$

If the curve passes through the origin, find the equation of the curve.

[Hint: Put $x^3 - 2x + 1 = u$.] ***(6 marks)***

Not for sale

HKCEE Questions (Level 1)

1. ***[A.Math 2009 Q.1(a)]***

Find $\int (4x+1)^2 \, dx$. ***(2 marks)***

5. ***[A.Math 2008 Q.1]***

Find $\int (8x+5)^{250} \, dx$. ***(2 marks)***

Not for sale

HKCEE Questions (Level 1)

3. ***[A.Math 2000 Paper II Q.1]***

Find $\int \sqrt{2x-1}\,dx$. ***(4 marks)***

4. ***[P.Math 2008 Paper II Q.5(a)]***

Find $\int y\sqrt{1-y^2}\,dy$. ***(2 marks)***

Not for sale

HKASL Questions (Level 1)

5. ***[A.Math 1980 Paper II Q.2]***

Find the indefinite integral $\int (x+2)\sqrt{x-1}\,dx$. ***(5 marks)***

6. ***[P.Math 2010 Paper II Q.5(a)]***

Find $\int (5-x)\sqrt{x-1}\,dx$. ***(2 marks)***

Not for sale

HKCEE Questions (Level 1)

7. ***[A.Math 2001 Q.2]***

Find $\int \frac{x}{\sqrt{3x^2+1}}\,dx$. ***(4 marks)***

8. ***[A.Math 1982 Paper II Q.1]***

Find the indefinite integral $\int \frac{x}{\sqrt{x+9}}\,dx$. ***(5 marks)***

Not for sale

HKDSE Questions (level 1)

9. ***[M2 2016 Q.7(a)]***

Using integration by substitution, find $\int (1+\sqrt{t+1})^2\,dt$. ***(4 marks)***

Not for sale

HKCEE Questions (Level 2)

10. ***[A.Math 1997 Paper II Q.5]***

The slope at any point (x, y) of a curve C is given by

$$\frac{dy}{dx} = 6x + \frac{1}{x^2}.$$

where $x > 0$. If the curve cuts the x-axis at the point (1, 0), find its equation.

(5 marks)

Not for sale

HKCEE Questions (Level 2)

11. ***[A.Math 1995 Paper II Q.1]***

The slope at any point (x, y) of a curve C is given by

$$\frac{dy}{dx} = 2x\sqrt{x^2+1}\,.$$

And C cuts the y-axis at the point (0, 1). Find the equation of C.

(5 marks)

Not for sale

HKCEE Questions (Level 2)

12. ***[A.Math 1994 Paper II Q.8]***

The slope at any point (x, y) of a curve C is given by

$$\frac{dy}{dx} = 8 - 10x .$$

and C passes through the point $A(1,13)$

(a) Find the equation of C.

(b) Find the equation of the normal to C at the point where C cuts the y-axis.

(7 marks)

Not for sale

HKCEE Questions (Level 2)

13. ***[A.Math 1999 Paper II Q.6]***

The slope at any point (x, y) of a curve C is given by

$$\frac{dy}{dx} = 3x^2 - 2x + k\ .$$

If the curve **touches** the x-axis at the point (2,0), Find

(a) the value of k,

(b) the equation of the curve. ***(6 marks)***

Not for sale

HKCEE Questions (Level 2)

14. ***[A.Math 1992 Paper II Q.4]***

The slope of the tangent of a curve C at any point (x, y) is $x^2 - 2$. C passes through the point (3, 4).

(a) Find an equation of C.

(b) Find the coordinates of the point on C at which the slope of the tangent is –2. ***(6 marks)***

HKCEE Questions (Level 2)

15. ***[A.Math 1993 Paper II Q.6]***

The slope of the tangent of a curve C at any point (x, y) is $3x^2 - 6x - 1$. C passes through the point (1, 0).

(a) Find an equation of C.

(b) Find the equation of the tangent to C at a point where C cuts the y-axis.

(7 marks)

Not for sale

HKDSE Questions (level 2)

16. ***[M2 2018 Q.5]***

(a) Using integration by substitution, find $\int x^3\sqrt{1+x^2}\,dx$

(b) At any point (x, y) on the curve Γ, the slope of the tangent to Γ is $15x^3\sqrt{1+x^2}$. The y-intercept of Γ is 2. Find the equation of Γ.

(7 marks)

Not for sale

HKDSE Questions (level 2)

17. ***[M2 2013 Q.4]***

The slope at any point (x, y) of a curve is given by $\frac{dy}{dx} = e^x - 1$. It is given that the curve passes through the point $(1, e)$.

(a) Find the equation of the curve.

(b) Find the equation of tangent to the curve at the point where the curve cuts the y-axis. ***(5 marks)***

Not for sale

HKCEE Questions (level 2)

18. ***[A.Math 2000 Paper II Q.6]***

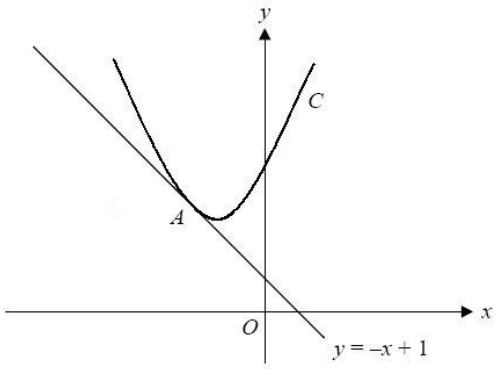


Figure 3

The slope at any point (x, y) of a curve C is given by $\frac{dy}{dx} = 2x + 3$.

The line $y = -x + 1$ is a tangent to the curve at point A. (See Figure 3.) Find

(a) the coordinates of A,

(b) the equation of C. ***(7 marks)***

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